Stock Market Integration Dynamics and its Determinants in the East Asian Economic Community Region

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JEL Classification: C13, C52, F21, F36, G15

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Abstract

In this paper, we examine the financial integration process of the equity market segments for the ASEAN+6 countries referred to in this study as East Asia Economic Community (EAEC) Region. The daily data for local equity indices and the Asian equity market benchmark is used from 1 January 1999 to 31st March 2015. Copula GARCH models have been employed to study the inter-temporal process of equity market integration. Empirical results show that the sample countries exhibit varying degrees of integration with the Asian benchmark. The 6 countries that form part of EAEC but not ASEAN (China, Japan, South Korea, Australia, India and New Zealand) exhibit high level of integration followed by ASEAN-5 (Indonesia, Malaysia, Philippines, Singapore and Thailand) members. Low integration level is observed for other ASEAN (Brunei, Cambodia, Laos, Myanmar and Vietnam) members which may be due to small size of their economies, lower trade integration and less developed capital markets. Equity portfolio flows within the EAEC region reconfirm the findings based on price data that regional integration is strengthening over time. Further, results from the panel data analysis show that trade integration and domestic capital market performance seem to be the fundamental drivers of financial integration in the EAEC region. The paper contributes to the International Finance literature, especially dealing with regional economic blocs by providing strong arguments for expanding the ASEAN Economic Community region in the near future to a more economically viable East Asian Economic Community region.

Keywords: stock market integration, ASEAN, time varying copula, panel data analysis, capital market development

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Section 1: Introduction

Economic globalisation and regional integration[1] are essential characteristics of the contemporary world economies. Rapid economic reforms around the world, characterized by liberalized financial sectors, have accelerated the process of globalization. International financial market liberalization has substantially raised the degree of capital mobility in the East Asian countries since 1990s (Guillaumin, 2009).The Asian countries have steered their way through the Asian Financial Crises by strengthening their macroeconomic fundamentals and improving their external conditions thereby promoting increasing cross-border trade and capital flows both within the region and with rest of the world. Deepening and widening of regional integration is paramount for increasing the macro prudential stability of the Asian economies. Financial Integration[2] shall provide benefit through more efficient allocation of capital, greater opportunities for risk diversification, better intertemporal consumption smoothing, a lower probability of asymmetric shocks and a more robust market framework (Pauer, 2005). Moreover, financial integration may also lead to financial development thereby contributing to economic growth in the region.[3] However; increasing financial linkages can trigger troublesome reversals of capital flow and contagion effects during crisis (Forbes and Rigobon, 2002). The key issue for both national and international policymakers is therefore not to choose between openness and autarky, but rather to formulate policies which help to maximize the longer-run gains of financial openness while minimizing the short-term risks.

Trade integration in Asia has steadily progressed over the last 20 years. Following the Asian crises, it was felt by the Asian policymakers that the financial integration in the region lagged the intra-regional trade integration and that liquid and well-regulated capital markets were quintessential for effective resource mobilization within the region. In the backdrop of the global financial crises, there was an increased realization in Asia to support economic rebalancing, thereby shifting focus from exports to developed countries to a more balanced economic structure supported by domestic and regional financial markets development. East Asian countries have accordingly strengthened their inter-governmental cooperation to promote regional financial integration. Their policy focus has shifted from a unilateral liberalized approach[4] to a bilateral and multilateral free trade area and investment agreements in the region (Hong Bum Jang, 2011). Much of the progress to date has been under the ASEAN[5] (Association of Southeast Asian Nations) and ASEAN+3 framework[5]- ASEAN Economic Community (AEC), Chiang Mai Initiative and Asian Bond Market Initiative. Another institution in the Asian context which is playing its part in the regional integration process in Asia is the Asian Development Bank (ADB). It is a regional development bank established on 19 December 1966 to promote social and economic development in Asia. ADB adopted the Regional Cooperation and International Strategy on 25 July 2006 thereby promoting monetary and financial integration as a focus of increasing regional cooperation and integration (ADB 2006). As global economic gravity shifts to east and 21st century is believed to be "The Asian Century", the policymakers of this region are convinced to accelerate the regional coordination so as to provide an efficient regional insurance arrangement[6] against the backdrop of global business cycle conditions. Also as Winkler (2010) points out that global financial crisis should act as a wakeup call for further progress on financial development and integration in the region. Given the increasing emphasis on closer regional co-operation across the globe, it has become important to study the current levels of financial integration within Asia. Particularly, the formation of ASEAN Economic Community (AEC) in the recently concluded 27th ASEAN summit in November 2015 could well be a rallying point for a pan-Asian integration in the years to come. Hence this study intends to evaluate the state of financial integration of the Asian countries specifically ASEAN+6 member countries[5] hereafter referred as East Asian Economic Community (EAEC) in the sphere of equity market segments using quantity based and price based approaches by applying contemporary econometric tools and techniques. Thus it aims to empirically evaluate the dynamic co-movements of these sample Asian equity markets with the benchmark Asian equity index which acts as proxy for a regional benchmark index. We further identify the fundamental determinants of these interrelationships which will help drive the process of integration with time. This paper also suggests policy roadmap to further deepen the financial development and integration in the region. The present study employs Copula Garch models on equity market price data and makes an important contribution to the existing literature by showing that the ASEAN+6 countries exhibit varying degrees of integration with the Asian benchmark. The data on equity portfolio flows reconfirms the increasing trend in regional integration for Asia. Further, trade integration and domestic capital market performance seem to be the fundamental drivers of financial integration in the region. We believe the present study would be instructive and complementary to the existing literature on financial integration and could provide strong arguments for expanding the ASEAN Economic Community region in the near future to a more economically viable East Asian Economic Community region. The empirical results presented in this paper have important implications for policy makers, portfolio managers and academia.

The rest of the paper is organized as follows: Section 2 deals with the trends and developments in the regional integration in Asia. Section 3 gives an overview of the review of literature. Section 4 presents the data while Section 5 covers the methodology related to Price based and Quantity based approach to Evaluate Equity Market Integration in the Asian Economic Community Region. Section 6 discusses the fundamental determinants of the equity market interdependence of this region while Section 7 provides summary and policy recommendations.

Section 2: Regional Integration in East Asian Economic Community-Trends and Recent Developments

Association of Southeast Asian Nations or ASEAN is the healthiest and most integrated regional organization in Asia comprising of diverse group of ten fast growing economies in different stages of economic and financial development. The ASEAN was established on 8 August 1967 in Bangkok with the signing of the ASEAN Declaration (Bangkok Declaration) by foreign ministers of Indonesia, Malaysia, Philippines, Singapore, and Thailand (henceforth referred as ASEAN-5). Since then, five new members have joined in i.e Brunei, Cambodia, Laos, Myanmar and Vietnam. The association has embarked on several economic integration[7] measures toward one vision, one identity and one community initiatives, including the ASEAN Free Trade Agreement (AFTA), the ASEAN Framework Agreement on Services (AFAS), and the ASEAN Investment Area (AIA). East Asian governments have embarked on policy initiatives for formal economic integration through bilateral and multilateral Free Trade Agreements (FTAs). The leaders of Japan, China and South Korea (henceforth referred to as Korea) were invited to the informal ASEAN Leaders' Meeting in December 1997, in the midst of the Asian financial crisis, which de facto initiated the ASEAN+3 process. In May 2000, the finance ministers of the ASEAN+3 agreed through the ''Chiang Mai Initiative'' to plan for closer monetary and financial cooperation. The Group also envisioned the progressive integration of the East Asian economies, ultimately leading to an "East Asian economic community".The idea of an East Asia Economic

Community was first mooted in the East Asia Vision Group (EAVG) Report of January 2001 entitled "Towards an East Asian Community: Region of Peace, Prosperity and Progress." In this report, the "integration of the East Asian economies, ultimately leading to an East Asia Economic Community" was envisaged and trade, investment and finance will be the catalysts in the community-building process. Specifically, the EAVG called for the establishment of the East Asia Free Trade Area (EAFTA) and the East Asia Investment Area (EAIA), among others. The first East Asian Summit (EAS) meeting was held in Kuala Lumpur in December 2005 and the second one in Cebu in January 2007, with the participation of ten ASEAN members and six countries including China, Japan, Korea, India, Australia, and New Zealand. East Asian Economic Community Region is a trade bloc for East Asian countries that may arise from ASEAN+3 or ASEAN+6 groups. Hence, this study makes reference to East Asian Economic Community as ASEAN+6 member group. Regional Comprehensive Economic Partnership (RCEP) is a proposed Free Trade Agreement between ASEAN and the 6 countries which together constitute the EAS group. Negotiations were formally launched in November 2012 at the ASEAN Summit in Cambodia. With this the RCEP will become the largest free-trade area with a total population of over 3 billion people (over 45% of world population), trade share estimated at around 27 per cent of global trade (based on 2012 WTO figures) and a combined GDP of around $US21 trillion (2013 IMF figures) i.e . around a third of the world's current annual GDP. Recently, the formation of the ASEAN Economic Community (AEC) by the ASEAN member states in the 27th ASEAN summit held in Kuala Lumpur, Malaysia in November 2015 is an important step toward regional economic integration in the Asian context. AEC will provide an economic region with smooth functioning regional financial system with more liberalized capital account regimes and interconnected capital markets facilitating greater trade and investment flows in the region.

For realizing capital market integration, some consistency in standards and connectivity in infrastructure is quintessential. ASEAN has made important strides in this area to reduce the costs of cross border trade of financial instruments leading to higher savings and investments in the region. The ASEAN Disclosure standards have enabled issuers to adopt uniform reporting practices while coming out with their equity/debt issues in the member states. The ASEAN Framework for cross border offering of collective investment scheme speeds up the process to authorize CIS managers to offer these products to retail investors in member states. The ASEAN

Trading Link launched in 2012, currently electronically links the stock exchanges of Malaysia, Singapore and Thailand. To further intensify the connectivity of the region's capital markets, ASEAN Capital Markets Infrastructure Blueprint was developed.

ASEAN has thus already played an important role to help promote Asia's integration. However, Asian integration process has moved slowly compared to rest of the world. Future economic cooperation in East Asia, leading to an East Asian Economic Community will broadly revolve around multiple agreements under the ASEAN+6 framework. While progress in East Asia has certainly been made on the regional trade front, a lot more needs to be done in the areas of investment and finance. As critical as trade is, economic integration is not based on trade alone, particularly in the context of international investment and capital market developments. Financial integration is also needed to strengthen regional economic integration.[8]

Section 3: Review of Literature

Financial integration is one of the most important topics in International Finance and thus, there is a wide range of studies focused on this issue. Economic theory and empirical findings suggest that financial integration and financial market development are likely to remove barriers to exchange, for allocating capital more efficiently and, thereby, contributing to economic growth (Calvi, 2010). Although there is a large body of literature on stock market integration around the world (Eun and Shim, 1989; Koch and Koch, 1991; Brocato, 1994; Francis and Leachman, 1998; Bessler and Yang, 2003) but the debate over degree of financial integration is still unsettled. Amongst the studies that suggest a lack of financial integration in the East Asian region, Danareksa Research Institute (2004) finds that financial integration in the region is still far behind that in Europe prior to its unification in the 1990s. Cavoli et al. (2004) show evidence of limited extent of financial market integration in East Asia. It finds out that the high income economies of Singapore, Hong Kong and Japan are highly integrated with the global markets than the low income economies of Thailand, Indonesia and Philippines. Kim et al. (2006) note that East Asian financial markets are relatively less integrated with each other than with the global markets. Lee (2008) finds that degree of regional financial integration in East Asia lags the trade integration. On the other hand, some studies have found a higher degree of financial integration in East Asia. Jeon et al. (2006) attribute the higher degree of financial integration in East Asia to the integration with the global market than with the regional counterparts. Chi et al.

(2006) finds high level of financial efficiency and integration in the East Asian equity markets. Using a Feldstein- Horioka[9] model, Guillaumin (2009) confirms a high degree of financial integration for select East Asian economies which have high incomes. Other recent studies (Yu et al., 2010; Park and Lee, 2011; Hinojales and Park, 2011; Park, 2013; Boubakri and Guillaumin, 2015) have documented that post 2008 crises, the regional integration in Asian economies has accelerated to mitigate world external shocks and ensure financial stability in the region.

The present study is based on two strands of literature. First set of literature is on how financial integration is measured. Two broad categories of measures of financial integration have been in vogue: price based and quantity based measures[10]. On the price based measures, there is a vast body of literature on the evaluation of financial integration using the traditional beta and sigma convergence (Adam et al., 2002; ECB, 2004; Rizavi et al., 2011). Early investigations into cross market linkages used the VAR models to study comovement of stock market returns. Contemporary literature has moved on to modern time series approaches i.e dynamic co-integration approach (Mylonidis and Kollias, 2010) and MGARCH models (BEKK model, DCC models and its extensions mainly Asymmetric DCC-GARCH of Cappiello et al., 2004 and Threshold DCC-GARCH of Pesaran et al., 2007) to study integration as a dynamic process rather than a realized one[11]. Recently, copula based models have been used to study the asymmetric nature of dependence between financial market returns (Patton, 2006; Rodriguez, 2007). The copula-GARCH (C-GARCH) model is a multidimensional GARCH process that models the interlinkages by using a copula function. The application of the C-GARCH model has recently attracted increased academic attention (Peng and Ng, 2011; Yang and Hamori, 2013; Yang et al. 2015). The quantity based measures assess the degree of integration based on cross country capital mobility and holding of assets and securities. For monitoring purposes, it is desirable to have indicators such as the price-based measures that are frequently available and supplement them by examining the quantity based measures to study the process of integration. The second set of literature is related to the fundamental determinants of stock market integration. Given the exploratory nature of the empirical investigation, we strive to include as many explanatory variables as possible after a thorough review of empirical literature on market integration (Refer table 1 for the variables and their reference study).

The paper fills important research gaps not covered by prior literature on Asian markets in the following ways: a) the existing literature covers only a subset of Asian countries whereas this paper studies the extent of integration for all of ASEAN+6 countries b) the studies on equity market integration in Asia are limited in their scope as they primarily employ the price based measures while we make use of the latest econometric methodology for the price based measures to evaluate the extent of integration and also support our analysis by quantity based measures c) the literature is virtually scanty when it comes to explaining the interrelationships found between the equity markets of the sample countries. We therefore survey the literature and try to identify the determinants of financial integration and suggest a suitable policy framework for achieving better equity market integration.

Table 1: Fundamental Determinants of Financial Integration

Category	Factor	Variable	Measurement	Reference Studies
Macro	Economic Growth & Development (EGD)	GDP growth rate	Annual Real GDP growth rate (%)	Vo and Daly, 2007; Guesmi et al., 2013; Ng et al., 2013
		GDP per capita	Real GDP per capita	Vo and Daly, 2007; Ng et al., 2013
		Index of Industrial Production	Log (IIP)	Buttner and Hayo, 2011; Guesmi et al., 2013
	External Position (EXP)	Current Account Balance	Current account as %of GDP	Guesmi et al., 2013
	Monetary Policy (MoP)	Inflation Rate	Annual Growth rate of GDP Deflator (%)	Vo and Daly, 2007; Huoy and Goh, 2007
		Short term Interest Rate	Repo rate (%)	Huoy and Goh, 2007; Guesmi et al., 2013
	Fiscal Position (FiP)	Budget Balance	Budget Balance/Surplus as %of GDP	Arfaoui and Abaoub, 2010
		Govt. Debt	Gross Public Debt as %of GDP	Gill, Sugawara and Zalduendo, 2014
		Govt. Tax Revenues	Tax Revenue as %of GDP	Vo and Daly, 2007
	Governance (GOV)	Governance Index	Arithmetic Mean of 6 Broad dimensions of governance	Vo and Daly, 2007; Ng et al., 2013
	Economic Risk (Ri)	Volatility of Forex Rate	EGARCH(1,1) based Conditional volatility estimate of FX Rate	Huoy and Goh, 2007; Guesmi et al., 2013
		Liquidity Risk	Money Supply (M2) to Total Reserves	Brunnermeier et al., 2008
Trade	Trade Openness (TO)	Trade Tariff	Trade Tariff as %ofDuty	
		Foreign Direct Investment	Net FDI as %of GDP	Ng et al., 2013
		Foreign Indirect Investment	Net FII as %of GDP	Ng et al., 2013
	Trade Linkages (TL)	Total Trade	Total Trade as %of GDP	Vo and Daly, 2007; Guesmi et al., 2013
		Regional Trade Intensity	Total Trade with the bloc members/ Total trade wih the world	Huoy and Goh, 2007
Market	Market Sophistication (MktS)	Market Size	Market Capitalization as %of GDP	Vo and Daly, 2007; Arfaoui and Abaoub, 2010
		Stock Market Efficiency	Stock Market Turnover Ratio	Vo and Daly, 2007; Narayan et al., 2014
	Market Performance (MktP)	Volatility of Market Return	EGARCH(1,1) based Conditional volatility estimate of market return	Abad and Chulia (2014)
		Domestic Growth Opportunities	PE ratio of the sample market index	Bekaert et al. (2007)
		Local Stock Market Conditions	Dividend Yield of the sample market index	Huoy and Goh, 2007; Guesmi et al., 2013

Section 4: Data

We begin our analysis on the daily stock market index closing prices retrieved from Bloomberg for ASEAN+6 countries for the period 1 January 1999 to 31st March 2015. The starting date has been selected keeping in mind the date of the Asian Crises period of 1997-1998[12]. All share market prices are obtained from a commonly used source, Morgan Stanley Capital International (MSCI), composed of stocks that broadly represent stock composition in the different countries. MSCI market price indices are free float-adjusted market capitalization indices, value weighted, and calculated with dividend reinvestment. Maghyereh (2004) has explained why the MSCI indices are better than other local stock indices[13]. As returns of the local equity indices would have been impacted by local currency movements against the US dollar, hence to capture the true market movements, all price indices are denominated in local currency. The periods of study as well as the selected equity indices differ only for four countries- Vietnam, Cambodia, Laos and Myanmar. Data for MSCI Vietnam Index was available only from 4th December 2006. MSCI indices were not available for Cambodia, Laos and Myanmar and hence the respective local benchmark equity indices were selected. Data for Laos Securities Exchange Composite Index was available from 12th January 2011 and Cambodia Stock Exchange Index was available from 19th April 2012. For Myanmar, Selective Myanmar Focusses Asia Index was used as market proxy, data for which was available from 20th November 2012. Thus, we have covered 15 countries in our study as no data is available for Brunei. The MSCI USA is included as a proxy for global factor as in prior research (Baele et al., 2004; Bartram, et al., 2007, Gupta et al., 2015). In addition, a pan-Asian index i.e MSCI ASIA, a benchmark index widely followed by investors who are investing in Asia, is used as a regional benchmark for Asia. The source for retrieving local, regional and global indices was also Bloomberg. We take the natural logarithm of the daily closing values. Daily returns are then computed as the first differences of the log-transformed series. As far as the quantity based indicators are concerned, we construct the annual portfolio equity inflows between each sample country and the ASEAN region as well the 6 countries which are part of EAEC and are outside the ASEAN for the period 2001 to 2014 hereafter referred as EAEC-ASEAN[14]. Similar portfolio exercise has been done for the portfolio equity outflows. The data for the same is sourced from IMF-CPIS (International Monetary Fund-Coordinated Portfolio Investment Survey) database. The quantity based approach has been applied for 11 countries as data was not available for Vietnam, Cambodia, Laos, Myanmar and Brunei. Further in case of China, portfolio outflow analysis has been performed using Hong Kong as a proxy as no data was available for mainland China. Finally, for the determinants of financial integration, an exhaustive set of factors were constructed (Refer table 1) belonging to 3 broad set of categories[15]i.e Macroeconomic, Trade and Market based category. The panel data analysis for identifying key determinants of stock market integration was again performed on the 11 countries for which quantity based indicators were employed. Data for the explanatory variables was not available for the remaining countries for the study period. The exhaustive list of variables surveyed from literature was sourced for all sample countries subject to their data availability from 1999 to 2014. These variables were predominantly of annual frequency except for volatility of forex rate, volatility of domestic market index return, domestic growth opportunities and local stock market conditions which were of daily frequency. The daily data was averaged over the year and it was not annualized as even a common scaling factor of 250 trading days would not have led to a change in its relative position across the sample.

Section 5: Assessing Stock Market Integration in East Asian Economic Community Region

Price Based Measures

The summary statistics for index returns series are reported in table 2.

Table 2: Descriptive Statistics of sample return series

Countries	Mean	Std. Dev.	Max	Min.	Skewness	Kurtosis	JB	LB	Obs.
INDON	0.0006	0.0183	0.1220	-0.1626	-0.1792	8.8531	5574* [0.00]	67.07* [0.00]	3891
MALAY	0.0003	0.0103	0.0697	-0.1024	-0.2714	11.7317	1204* [0.00]	93.42* [0.00]	3776
PHILIP	0.0002	0.0145	0.1629	-0.1367	0.3042	14.9823	2382* [0.00]	75.01* [0.00]	3973
SING	0.0002	0.0129	0.1043	-0.0983	-0.0879	8.6729	5321* [0.00]	27.88* [0.00]	3965
THAI	0.0003	0.0177	0.1586	-0.1808	-0.0023	12.7920	1482* [0.00]	43.34* [0.00]	3712
CAMB	-0.0015	0.0147	0.0486	-0.0513	-0.4146	6.4738	260* [0.00]	29.27* [0.00]	491
LAOS	0.0003	0.0137	0.0486	-0.0502	0.1728	5.9154	321* [0.00]	69.29* [0.00]	896
MYAN	0.0003	0.0089	0.0286	-0.0436	-0.4186	5.2566	119* [0.00]	15.53 [0.17]	494
VIET	0.0000	0.0187	0.0879	-0.0906	-0.0052	4.5347	192* [0.00]	140.97* [0.00]	1958
CHINA	0.0002	0.0192	0.1406	-0.1284	0.0845	8.3641	4768* [0.00]	51.85* [0.00]	3973
JPN	0.0000	0.0142	0.1306	-0.1044	-0.3017	9.0475	5983* [0.00]	42.38* [0.00]	3888
KOR	0.0005	0.0186	0.1172	-0.1309	-0.2124	7.2554	2878* [0.00]	14.28 [0.28]	3777

AUS	0.0002	0.0105	0.0610	-0.0868	-0.4023	8.7292	5473* [0.00]	33.49* [0.00]	3925
INDIA	0.0004	0.0165	0.1642	-0.1205	-0.1504	9.7384	7108* [0.00]	48.14* [0.00]	3750
NZ	0.0000	0.0104	0.0614	-0.0611	-0.1919	5.9185	1388* [0.00]	38.12* [0.00]	3846
ASIA	0.0001	0.0127	0.0969	-0.0884	-0.2684	7.3224	3140* [0.00]	29.95* [0.00]	3973
USA	0.0001	0.0129	0.1104	-0.0951	-0.0862	10.2277	8652* [0.00]	54.68* [0.00]	3973

Notes: * denotes level of significance at 5% . Values in parentheses [ ] indicate the p-values.
b.) JB=JarqueBera and LB= Ljung Box.LB statistics is reported up to 12 lags.
c). INDON, MALAY, PHILIP, SING, THAI, VIET, CHINA, JPN, KOR, AUS, INDIA, NZ, ASIA, USA denote MSCI Indonesia, MSCI Malaysia, MSCI Philippines, MSCI Singapore, MSCI Thailand, MSCI Vietnam, MSCI China, MSCI Japan, MSCI South Korea, MSCI Australia, MSCI India, MSCI New Zealand, MSCI Asia and MSCI USA indices. CAMB, LAOS and MYAN denote local market indices. Data for all the indices are in local currency
d.) Source: Bloomberg

The annualized mean returns[16] for sample countries are positive for all except Cambodia highest being 12.5% for Korea. Annualized volatility in returns[16] ranged between 14.07% to 30.41% (highest being for China and the lowest for Myanmar). The low volatility in returns for Myanmar could be attributed to its virtually dormant equity market[17]. The skewness statistic for majority of the sample markets was negative and all of them show a clear excess kurtosis thereby implying fat tails and sharp peaks. The Jarque Bera multiplier for all markets is statistically significant, thereby indicating that the return distributions are non-normal. The results of Ljung-Box Q-statistics up to 12 lags in levels clearly state the presence of serial correlation for all sample return series except Korea and Myanmar. As a pre-cursor to the time-series analysis, we also conducted the unit root tests for stationarity. The Augmented Dickey–Fuller (ADF) as well as the Phillips–Perron test results[18] suggest that all sample series are integrated to order 1 i.e I(1). Table 3 gives the Pearson, Kendall’s tau and Spearman’s rho correlations between each index return of the sample country and the Asian benchmark index return. Because Pearson correlation coefficients test for linear association, they are neither appropriate for indicating non-linear relationships nor robust for distributions with long tails. Similarly, nonparametric rank correlations tests, such as Kendall's tau and Spearman's rho, are relatively insensitive to the observations at the tails.

Table 3: Correlation estimates of dependence between sample country returns with Asian benchmark returns

	Pearson's Correlation	Kendal's Tau	Spearman's Rho
INDON	0.427	0.259	0.371
MALAY	0.414	0.28	0.399
PHILIP	0.395	0.231	0.336
SING	0.589	0.378	0.528
THAI	0.436	0.272	0.392
CAMB	0.047	0.027	0.039
LAOS	-0.074	-0.053	-0.077
MYAN	-0.015	-0.019	-0.028
VIET	0.222	0.119	0.174
CHINA	0.628	0.418	0.576
JPN	0.886	0.68	0.857
KOR	0.619	0.443	0.609
AUS	0.621	0.386	0.542
INDIA	0.405	0.249	0.359
NZ	0.308	0.171	0.251

Notes: The table summarizes the parametric and non-parametric tests for correlation

The results (refer Table 3) clearly show that Japan has the highest correlation with the Asian benchmark index for all the three different measures of correlations. The average correlation coefficients for ASEAN and ASEAN-5 member states with the Asian benchmark were lower than the average EAEC-ASEAN group members for all the three measures of correlations. The highest correlation amongst the ASEAN member states with the Asian benchmark is that for Singapore which is lower than 4 out of the 6 members of EAEC-ASEAN group (China, Japan, Korea, and Australia). Our findings imply that in general the association of EAEC-ASEAN group is higher with the Asian benchmark compared to that of the ASEAN group.

5.1 Methodology

Multivariate normality assumption is not suitable for measuring the dependence structure of equity returns between two markets, especially with respect to their asymmetric co-movements or contagion effects (Longin&Solnik, 2001; Poon et al., 2004). A copula-based measure can specify the dependence structures while accounting for non-linearity without the constraint of normality. Also, based on Sklar's Theorem (Sklar, 1959) which helps in separating the marginal and joint distributions, the copula method makes the estimation process more flexible. Patton (2006) extended this methodology by adding a time-varying specification to capture the dependence over time. Let $R_{countryi,t}$ and $R_{asia,t}$ be random variables that denote country i's stock index returns and Asian benchmark index returns respectively at period t, with marginal conditional cumulative distribution function $U_{countryi,\ t} = G_{countryi,t}\ (R_{countryi,t}|\phi_{t-1})$ and $U_{asia,\ t} = G_{asia,t}\ (R_{asia,t}|\phi_{t-1})$, where ϕ_{t-1} denotes past information. Then the conditional copula function $C_t\ (U_{countryi,t}, U_{asia,t}\ |\phi_{t-1})$ can be written using the two time varying cumulative distributive functions. Extending Sklar's theorem, bivariate conditional cumulative distributive function of random variable $R_{countryi,t}$ and $R_{asia,t}$ is:

$$\mathbf{F\ (R_{countryi,t}, R_{asia,\ t}\ |\ \phi_{t-1}) = C_t\ (U_{countryi,t}, U_{asia,\ t}\ |\ \phi_{t-1})}$$

5.1.1 Marginal Specification

We investigate the integration of each country's stock index with the Asian benchmark index by combining the copula functions, described above, with a GARCH-type model of conditional heteroscedasticity. In this paper, we adopt EGARCH(1,1) model (Nelson,1991) for specification of conditional volatility to capture the asymmetric impacts of shocks or innovations on volatilities and to avoid imposing nonnegativity restrictions on the values of GARCH parameters. EGARCH(1,1) model is chosen as the preferred model in the interest of parsimony of parameters (see Kim and Wang, 2006). The mean equation used for country i's returns and Asian benchmark returns is as follows:

$$\mathbf{R_{countryi,t,} = \mu + R_{countryi,t-1} + R_{usa,t-1} + \varepsilon_t}$$

If $\varepsilon_t = \sigma_t z_t$ where z_t is standard Gaussian, then EGARCH (1,1):

$$\mathbf{Ln\ (\sigma^2_t) = \omega + \alpha\ (|z_{t-1}| - E\ [|z_{t-1}|]) + \Upsilon\ z_{t-1} + \beta\ Ln\ (\sigma^2_{t-1})}$$

Here, R_{usa} is the return of the USA index and is used as a global factor. It is well known that many financial time series are non-normal and tend to have a fat tail behavior (Mandelbrot, 1963; Fama, 1965). Thus a GARCH is formulated after factoring in the mean spillover effects from the US which proxies for global effects (see Dungey, Fry, & Martin, 2003, Samitas & Kenourgios, 2007). The lagged US returns proxy for global factor and capture the global mean spillover effects. In order to better capture the skewed and fat tails characteristic, Bollerslev

(1987) suggests the use of the conditional Student-t distribution for the error term. We further obtain the standardized residuals from this marginal specification.

5.1.2 Copula Models for Dependence

Having estimated each of the marginal distribution, we now apply the probability integral transform to convert the standardized residuals of the marginal distribution into Uniform (0, 1) distributions. To describe the symmetric and asymmetric dependence structure between the uniform distributions of each country with that of the Asian benchmark, the study uses two Elliptical (Gaussian and Student t) and two Archimedean's (Clayton and Gumbel) family of copula models. Because the nature of copulas can be either static or time-variant, hence we include both for our empirical analysis.

Static Copula Models

a. Gaussian Copula: Following (Patton, 2006), the dependence parameter of Gaussian process is

$$\mathbf{C_{Ga}\ (u,v;\rho)} = \int_{-\infty}^{\phi^{-1}(\boldsymbol{u})} \int_{-\infty}^{\phi^{-1}(\boldsymbol{v})} \frac{\mathbf{1}}{\mathbf{2\pi\sqrt{1-p^2}}} \mathbf{exp}\ \Big(-\ \frac{x_1^2-2\rho x_1\ x_2+x_2^2}{2(1-\rho^2}\Big)\ \mathbf{d}\boldsymbol{x_1}\ \mathbf{d}\boldsymbol{x_2}$$

$$= \phi_p\ (\phi^{-1}(u), \phi^{-1}(v);\ \rho)$$

Where u and v are cumulative distribution functions of standardized residuals, subjected to a uniform distribution between 0 and 1, ρ is Pearson's linear correlation,$\phi^{-1}$is the inverse cumulative distribution function of a standard normal distribution.

b. T- Copula: The dependence parameter of t copula follows from (Fernandez, 2008):

$$\mathbf{C\ (u,v)} = \int_{-\infty}^{T_v^{-1}(\boldsymbol{u})} \boldsymbol{dx} \int_{-\infty}^{T_v^{-1}(v)} \boldsymbol{dy} \frac{\mathbf{1}}{\mathbf{2\pi\sqrt{1-p^2}}} \left[\mathbf{1} + \frac{x^2-2\rho xy+y^2}{v(1-\rho^2}\right]^{-\frac{v+2}{2}}$$

$$T_v(\mathrm{x}) = \int_{-\infty}^{x} \frac{\Gamma\ ((\frac{v+1}{2}))}{\sqrt{\pi v}\Gamma\ (\frac{v}{2})} \left(1+\frac{z^2}{v}\right)^{-\frac{v+1}{2}} \mathrm{dz}$$

T is the student t distribution with degrees of freedom v and Pearson's correlation ρ. In comparison to Gaussian copula, t copula captures the tail dependence

c. Gumbel Copula: The dependence parameter (Gumbel, 1960) which can capture the upper tail dependence:

$$\mathbf{C_{Ga}\ (u,v;\boldsymbol{\theta}) = exp}\ (-((-\mathbf{ln\ u})^{\boldsymbol{\theta}} + (-\mathbf{ln\ v})^{\ \boldsymbol{\theta}}\ )^{\ \frac{1}{\theta}})$$

Where $1 \leq \theta < +\infty$; upper limit in Gumbel copula is ∞

d. Clayton Copula: The dependence parameter(Clayton, 1978) captures the lower tail dependence:

$$\mathbf{C_{CL}\ (u,v;\boldsymbol{\theta}) =\ (\boldsymbol{u}^{-\boldsymbol{\theta}} + \boldsymbol{v}^{-\boldsymbol{\theta}} - 1)^{-1/\boldsymbol{\theta}}}$$

Where $0 \leq \theta < +\infty$;

Time Varying Copula Models

Time varying copulas can be considered as dynamic generalization of a Pearson correlation or Kendall's tau but it is difficult to find causal variables to explain such characteristics (Patton, 2006). In practice though time varying copulas are operationalized to follow autoregressive moving average (ARIMA) (p,q) process.

a. Time varying Gaussian Copula (Patton, a2006) uses a coefficient ρ_t to study the dependence dynamics defined as:

$$\boldsymbol{\rho_t} = \widetilde{\Lambda}\ (\boldsymbol{\omega_N} + \boldsymbol{\beta_{N\,1}}.\ \rho_{t-1} + \ \ldots\ldots + \boldsymbol{\beta_{N\,p}}.\ \rho_{\ t-p} + \boldsymbol{\alpha_N}\ .\ \frac{1}{q}\sum_{j=1}^{q} \Phi^{-1}\ \boldsymbol{v_{t-j}}))$$

Where $\widetilde{\Lambda}$ (x) is a logistic transformation which is defined as $\widetilde{\Lambda}$ (x) = (1-e^{-x})(1-e^{-x})$^{-1}$

b. Time varying T copula (Patton, b2006) uses the following to study the dependence process:

$$\boldsymbol{\rho_t} = \widetilde{\Lambda}\ (\boldsymbol{\omega_T} + \boldsymbol{\beta_{T\,1}}.\ \rho_{t-1} + \ldots + \boldsymbol{\beta_{T\,p}}.\ \rho_{\ t-p} + \boldsymbol{\alpha_T}\ .\ \frac{1}{q}\sum_{j=1}^{q} \boldsymbol{T^{-1}}\ (\boldsymbol{u_{t-j}}\,;\boldsymbol{DoF}).\boldsymbol{T^{-1}}(\boldsymbol{v_{t-j}}\,;\boldsymbol{DoF}))$$

Where T^{-1} is the inverse function of the student t-distribution with given degrees of freedom (DoF)

c. Time varying Gumbel copula studies the dependence using θ_t corresponding to $\tau_t = 1 - 1/\theta_t$ defined as:

$$\boldsymbol{\tau_t} = \Lambda\Big(\boldsymbol{\omega_G} + \boldsymbol{\beta_{G\,1}}.\boldsymbol{\tau_{t-1}} + \ldots + \boldsymbol{\beta_{G\,p}}.\boldsymbol{\tau_{t-P}} + \boldsymbol{\alpha_G}\,.\frac{1}{q}\sum_{j=1}^{q} |\boldsymbol{u_{t-j}} - \boldsymbol{v_{t-j}}\,|\Big)$$

Where $\Lambda(\mathrm{x}) = (1 + e^{-x})^{-1}$

d. Time varying Clayton copula uses θ_t corresponding to $\tau_t = \theta_t/(2 + \theta_t)$ to find the dependence defined as:

$$\boldsymbol{\tau_t} = \Lambda\big(\boldsymbol{\omega_C} + \boldsymbol{\beta_{C\,1}}.\boldsymbol{\tau_{t-1}} + \ldots + \boldsymbol{\beta_{C\,p}}.\boldsymbol{\tau_{t-P}} + \boldsymbol{\alpha_{C1}}.\,|\boldsymbol{u_{t-1}} - \boldsymbol{v_{t-1}}\,| + \ldots + \boldsymbol{\alpha_{Cq}}\,.\,|\boldsymbol{u_{t-q}} - \boldsymbol{v_{t-q}}\,|\big)$$

The Elliptical copula models (Gaussian and Student t) are most popular in Finance literature due to the ease with which they can be implemented. Gaussian copula is symmetric and has no tail dependence whereas Student t copula can capture extreme dependence between variables[19]. The Gumbel copula exhibits greater dependence in the upper tail than in the lower, whereas the Clayton copula exhibits greater dependence in the negative tail than in the positive tail.

5.1.3 Estimation of Copula Parameters

Parameters are normally estimated in two steps, first for the marginal and second for the copula. The approach is similar to the Inference function for Margin (IFM) method. IFM method is superior to Exact Maximum Likelihood (EML) as the latter needs expensive computation more so for higher dimensions. In IFM method, parameters of the marginal distributions can be estimated before those of the copula functions whereas the EML method requires both to be estimated simultaneously.

5.1.4 Goodness of Fit Tests

Besides the log likelihood ratios, in order to check the quality of the overall fit of the family of copula models, goodness of fit test of (Genest et al. ,2009) which is based on a comparison of the distance between the estimated and the empirical copula: $C_n = \sqrt{n}\ (C_n - C_{\theta_n})$.

The test statistic is based on Cramer-Von Mises distances defined as: $S_n = \int C_n\ (u)^2 \mathbf{d}C_n\mathbf{(u)}$

Large values of the statistic S_n leads to rejection of the null hypothesis that the Copula C belongs to the class C_0. A multiplier approach is used to find the p-values associated with the test statistics (Kojadinovic and Yan, 2011). The highest p-value indicates that the distance between the empirical and the estimated copulas is the smallest and that the copula in use provides the best fit (Aloui et al., 2013).

5.2 Empirical Results

5.2.1 Price Based Measures

Table 4 summarizes the results of the marginal distribution. The coefficient of the lagged US returns acting as a proxy for global factor to capture mean spillover effect was found to be significant for all the countries except for Cambodia, Laos and Myanmar which can be attributed to their low level of economic globalization and less active equity markets.

Table 4: Marginal Specification of Stock Returns

	Mean Eqn			Variance Eqn			
Country	**μ**	**$R_{country\ I}$ (-1)**	**R_{USA} (-1)**	**ω**	**α**	**ϒ**	**β**
INDON	0.0006* (0.00)	0.0412* (0.02)	0.3616* (0.02)	-0.5915* (0.00)	0.2614* (0.00)	-0.0714* (0.00)	0.9515* (0.00)
MALAY	0.0002* (0.00)	0.0742* (0.01)	0.2131* (0.01)	-0.2747* (0.03)	0.1988* (0.02)	-0.0437* (0.00)	0.9865* (0.00)
PHILIP	0.0002 (0.00)	0.0845* (0.01)	0.3726* (0.01)	-1.0235* (0.15)	0.2727* (0.03)	-0.0593* (0.02)	0.9067* (0.02)
SING	0.0002 (0.00)	-0.0689* (0.02)	0.3408* (0.01)	-0.2337* (0.03)	0.1753* (0.02)	-0.0481* (0.01)	0.9892* (0.00)
THAI	0.0003 (0.00)	0.0442* (0.01)	0.3025* (0.02)	-0.4579* (0.06)	0.2148* (0.02)	-0.0451* (0.02)	0.9648* (0.01)
CAMB	-0.0001 (0.00)	0.0059 (0.04)	0.0084 (0.02)	-0.4369* (0.07)	5.1756* (1.97)	-0.1452 (0.33)	0.9788* (0.01)
LAOS	-0.0001 (0.00)	-0.0967* (0.03)	0.0009 (0.02)	-1.5136* (0.31)	0.7318* (0.15)	-0.0426 (0.05)	0.8703* (0.03)
MYAN	0.0006 (0.00)**	0.0011 (0.04)	-0.0065 (0.04)	-0.5815* (0.25)	0.1669* (0.06)	-0.0464 (0.04)	0.9519* (0.02)
VIET	-0.0004 (0.00)	0.1723* (0.00)	0.2680* (0.02)	-0.5951* (0.10)	0.3067* (0.00)	-0.0388* (0.02)	0.9565* (0.01)
CHINA	0.0002 (0.00)	0.0111 (0.01)	0.4911* (0.02)	-0.2024* (0.03)	0.1457* (0.01)	-0.0467* (0.00)	0.9886* (0.01)
JPN	0.0001 (0.00)	-0.0183 (0.02)	0.4726* (0.01)	-0.4525* (0.06)	0.1839* (0.09)	-0.0736* (0.01)	0.9651* (0.01)
KOR	0.0003** (0.00)	-0.0374* (0.01)	0.4834* (0.02)	-0.1758* (0.02)	0.1478* (0.01)	-0.0435* (0.01)	0.9927* (0.00)
AUS	0.0003* (0.00)	-0.0819* (0.01)	0.3901* (0.01)	-0.2452* (0.03)	0.1289* (0.01)	-0.0664* (0.01)	0.9853* (0.00)
INDIA	0.0006* (0.00)	0.0420* (0.02)	0.2383* (0.02)	-0.4219* (0.05)	0.2234* (0.02)	-0.0885* (0.01)	0.9708* (0.00)
NZ	0.0001 (0.00)	-0.0134 (0.01)	0.2504* (0.01)	-0.1259* (0.02)	0.0784* (0.01)	-0.0277* (0.01)	0.9931* (0.00)
ASIA	4.97 (0.00)	-0.0429* (0.01)	0.4862* (0.01)	-0.2938* (0.04)	0.1632* (0.0168)	-0.0566* (0.01)	0.9819* (0.00)

Notes: Figures in ( ) includes standard error. */** indicates significance at 5%/10% level of significance
b.)$R_{countryi,t}$ and $R_{asia,t}$ denote country i's stock index returns and Asian benchmark index returns respectively.
c.) Coefficient ω measures the constant term; Coefficient ϒ measures leverage effect while Coefficient β measures persistence (GARCH effect)

The EGARCH (1,1) estimation shows that long run volatility persistence (measured by coefficient β) was statistically significant for all the markets. The asymmetric effect of news on volatility factor (measured by coefficient ϒ) is negative for all countries and even statistically significant (except for smaller ASEAN member states i.e Cambodia, Laos and Myanmar) thereby justifying the use of EGARCH (1,1) model. The ACFs of the standardized residuals and squared standardized residuals, obtained from the mean equation but not presented due to brevity of space, indicate that the standardized residuals are approximately i.i.d. They are therefore more suitable to copula estimation than the raw return series. We turn to estimate the copula functions wherein firstly we consider the standardized residuals and transform it into vector of uniform variates using cumulative distributive function.

Table 5: Estimation Results of Unconditional Copula Models

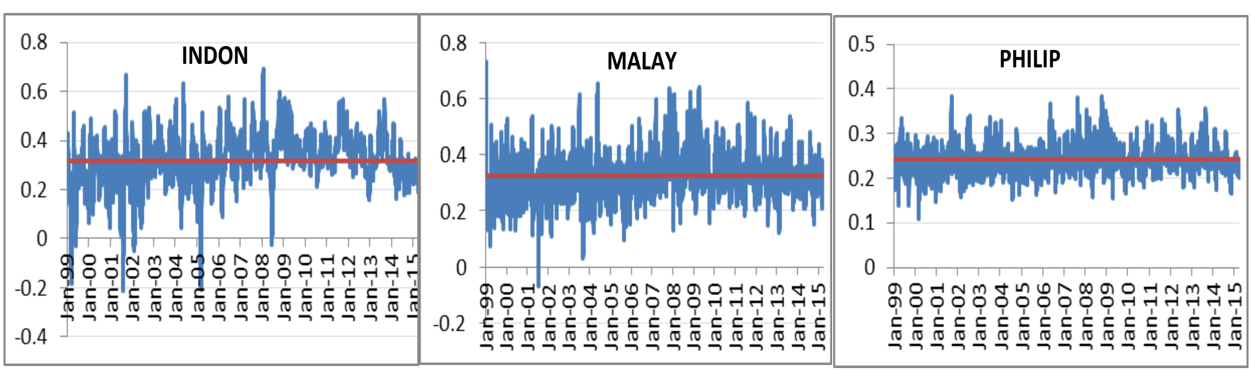
	Gaussian		**Student t**		**Clayton**		**Gumbel**	
	Value	**LL**	**Value**	**LL**	**Value**	**LL**	**Value**	**LL**
INDON	0.32	-205.19	**0.32**	**-218.01**	0.39	-179.35	1.22	-178.08
MALAY	0.32	-208.67	**0.32**	**-223.44**	0.41	-187.61	1.23	-183.68
PHILIP	0.24	-119.29	**0.24**	**-121.91**	0.29	-107.97	1.15	-91.22
SING	0.48	-521.78	**0.48**	**-548.91**	0.73	-476.26	1.41	-457.25
THAI	0.34	-228.57	**0.34**	**-237.45**	0.43	-197.03	1.25	-198.31
CAMB	0.00	0.00	**0.00**	**-0.36**	0.00	0.00	1.10	0.00
LAOS	-0.08	-3.12	**-0.08**	**-3.19**	0.00	-0.01	1.10	-2.58
MYAN	**0.02**	**-0.11**	0.02	-0.10	0.00	0.00	1.11	0.00
VIET	0.09	-8.66	**0.08**	**-14.83**	0.13	-13.44	1.12	-1.06
CHINA	0.51	-602.35	**0.52**	**-628.30**	0.74	-486.99	1.47	-555.18
JPN	0.83	-2260.70	**0.83**	**-2343.80**	2.29	-1913.10	2.50	-2141.80
KOR	0.56	-710.32	**0.57**	**-744.09**	0.89	-602.31	1.55	-648.12
AUS	0.43	-404.99	**0.43**	**-431.08**	0.63	-381.48	1.35	-352.03
INDIA	0.33	-210.62	**0.33**	**-213.14**	0.37	-155.86	1.23	-188.88
NZ	0.13	-32.54	**0.13**	**-39.41**	0.16	-39.07	1.11	-19.11

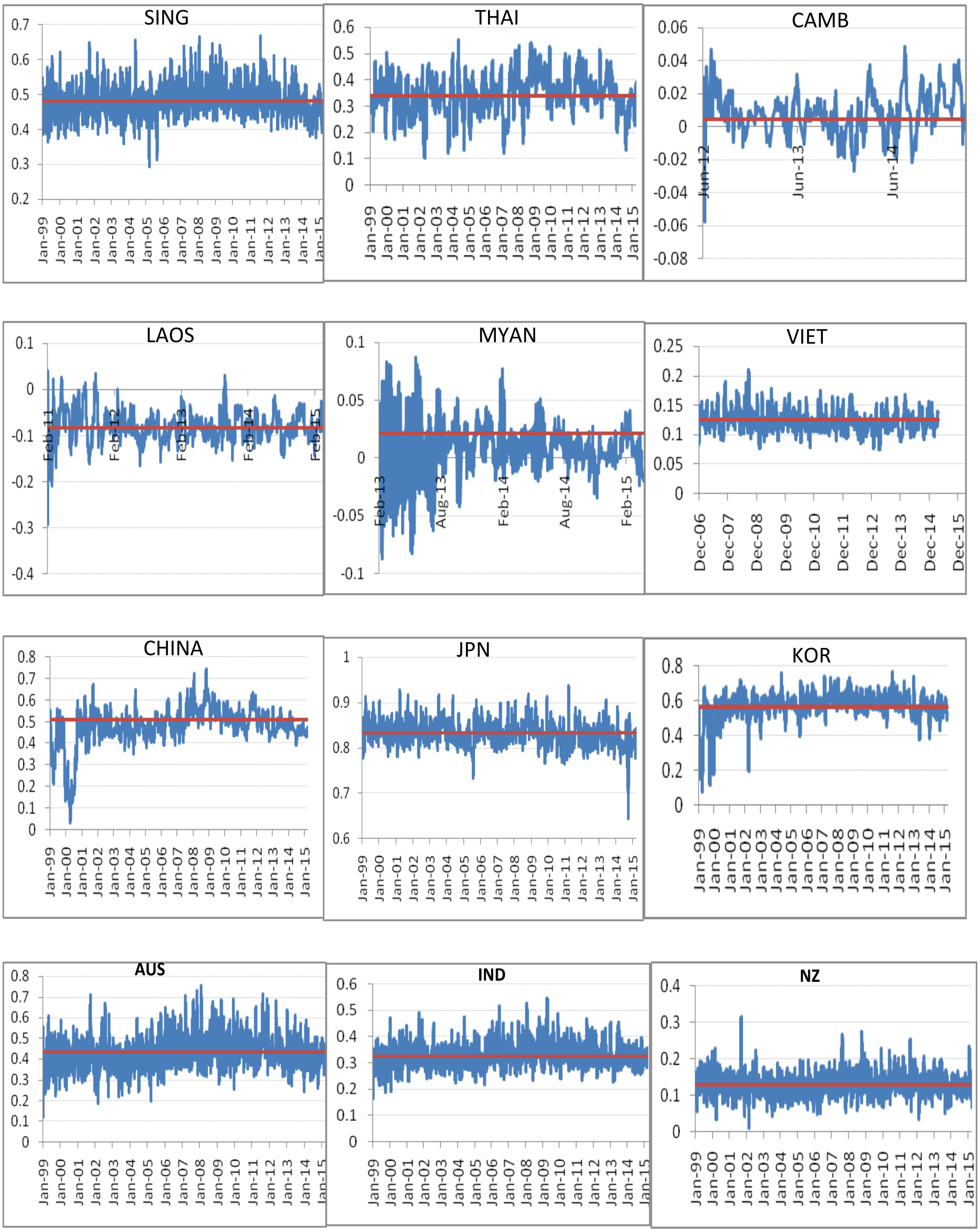
Notes: Value denotes the dependence parameter between the respective country stock returns and the Asian benchmark index returns
b.) LL denotes the log likelihood value of the dependence parameter between the respective country stock returns and the Asian benchmark index returns using the different copula models
c.) Value in bold denotes lowest LL values thereby providing the best fit copula model amongst the family of models under study

Results from the unconditional copula models (Refer table 5) indicate that irrespective of the choice of copula function, the dependence parameter of the Japanese stock returns is the highest (0.83) with the Asian benchmark stock returns. Amongst the ASEAN member states, highest integration level is exhibited by Singapore (with a dependence parameter of 0.48) which also has the highest trade to GDP ratios amongst all the Asian countries over the study period (with latest figures of 350.85% as on December 2014, Source World Bank). Other ASEAN-5 member states are having a dependence parameter of around 0.3 with the exception of Philippines (0.24). Amongst the EAEC-ASEAN group members, New Zealand has the lowest dependence parameter (0.13) with the Asian benchmark. China, Japan and Korea are respectively much more integrated (having a dependence measure more than 0.50) than other sample countries in the Asian region. The average measure of dependency for the ASEAN-5 group countries with the Asian benchmark is 0.3416 while the same for ASEAN group is 0.1928. The average value of dependence parameter for the EAEC-ASEAN group members is 0.4674. Bekaert et al. (2005) and Goetzmann, Li, and Rouwenhorst (2005) claimed that capital market integration and increased trade are embedded with predictions about the association between markets. The copula functions have positive parameters for all countries except Laos which implies that all sample country's stock index returns besides that of Laos correlates positively with the Asian benchmark. Also the copula parameter is close to 0 for Cambodia, Myanmar and Vietnam which can be attributed to their relatively small and inactive equity markets. For the unconditional copula models, T copula seems to be the best fit (lowest log likelihood values, refer table 5) amongst the copula family of models for all countries except Myanmar for which normal copula fits the data.

Results from the conditional copula models (Refer Figure 1) provide a time varying dependence measure between each sample country's stock index returns and the Asian benchmark index returns.

Figure 1: Conditional Copula Models- Dependences between sample country returns with Asian benchmark returns
Source: Authors calculations




SING
THAI
CAMB
LAOS
MYAN
VIET
CHINA
JPN
KOR
AUS
IND
NZ

We have charted the best fit conditional copula model so selected from the goodness of fit tests (Refer table 6) against its corresponding unconditional copula parameter to study its inter temporal dependence.

Table 6: Goodness of Fit Test for different copula models

	Gaussian	**Student t**	**Clayton**	**Gumbel**
INDON	0.0063	**0.0125**	0.0000	0.0000
MALAY	**0.0068**	0.0004	0.0005	0.0006
PHILIP	**0.1074**	0.0008	0.0004	0.0003
SING	**0.0161**	0.0009	0.0004	0.0008
THAI	**0.3332**	0.0010	0.0001	0.0005
CAMB	**0.7757**	0.5779	0.7313	0.5376
LAOS	**0.4790**	0.1154	0.3352	0.2150
MYAN	0.0305	0.0095	0.0000	0.0000
VIET	0.1763	0.0550	**0.3731**	0.0150
CHINA	**0.0110**	0.0000	0.0000	0.0000
JPN	0.0014	**0.0065**	0.0000	0.0000
KOR	**0.0067**	0.0006	0.0005	0.0002
AUS	**0.0120**	0.0000	0.0000	0.0000
INDIA	**0.1120**	0.0010	0.0000	0.0000
NZ	**0.0265**	0.0005	0.0214	0.0015

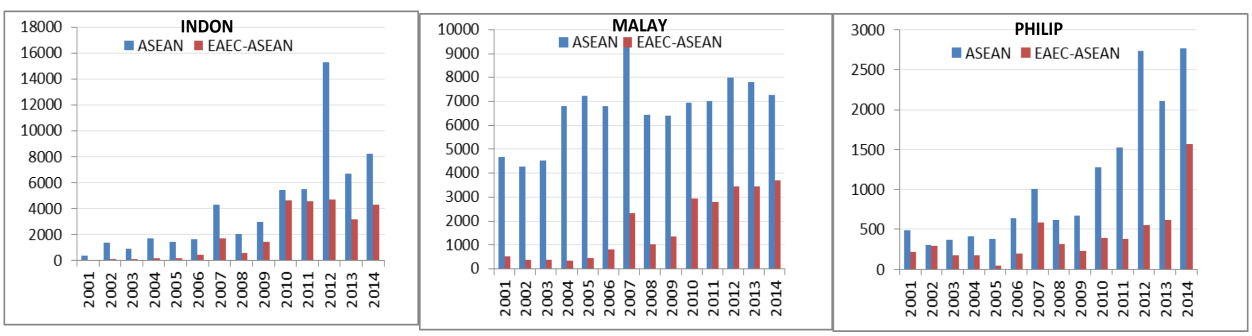
Notes: The table presents the p-values of the goodness-of-fit test. Large p-values (bold face numbers) indicate that the copula provides best fit to the data.

Stronger co-movements between the sample markets and Asian benchmark index reflect higher levels of integration and hence, an increase in dependence parameter (as a measure of co-movement) among financial markets may signal increased convergence (Kuper&Lestano, 2007). Within the ASEAN-5 basket, the highest time varying dependence with the Asian benchmark index was shown by Singapore market while the lowest was for Philippines. The standard deviation of these time varying dependences for the ASEAN-5 members was highest for Indonesia and lowest for Philippines. As for the EAEC-ASEAN group members, Japan showed the highest time varying dependence and also the lowest variance of the dependence over time and thus seems to be having highest levels of equity market integration with the Asia. Although China and Korea seem to have higher average level of dependences over time with the Asian benchmark index but the former seems to have a larger variance in these associations. New Zealand seems to have the lowest level of equity market integration which is slightly higher than the 4 newer ASEAN members (Cambodia, Laos, Myanmar and Vietnam).Amongst the 4 new ASEAN members, Laos has the lowest levels of integration which is even negative for majority of its sample period under study. Vietnam shows the highest level of equity market integration amongst these 4 countries which on an average basis is equivalent to the integration levels of New Zealand.

5.2.2 Quantity Based Measures

Quantity based measures supplement the price based measures for assessing and monitoring the changes and trends in the equity market integration. Investors in general seem to allocate a disproportionately large part of their equity holdings in domestic stocks, a phenomenon known in academic literature as “home bias” (Tesar and Werner, 1992, and Lewis, 1999). International diversification can be assumed to be increasing with decreasing home bias. Thus we examine the cross border equity holdings of the sample Asian countries to assess whether there is a shift to investment within/from ASEAN region or the broad ASEAN+6 region. Figure 2 shows the annual values of portfolio equity inflows from the ASEAN and EAEC-ASEAN groups into the sample countries respectively.

Figure 2: Portfolio Equity Inflows into sample Asian countries from ASEAN and EAEC-ASEAN groups (all figures are end of period and in millions USD)
Source: IMF CPIS, Authors calculations



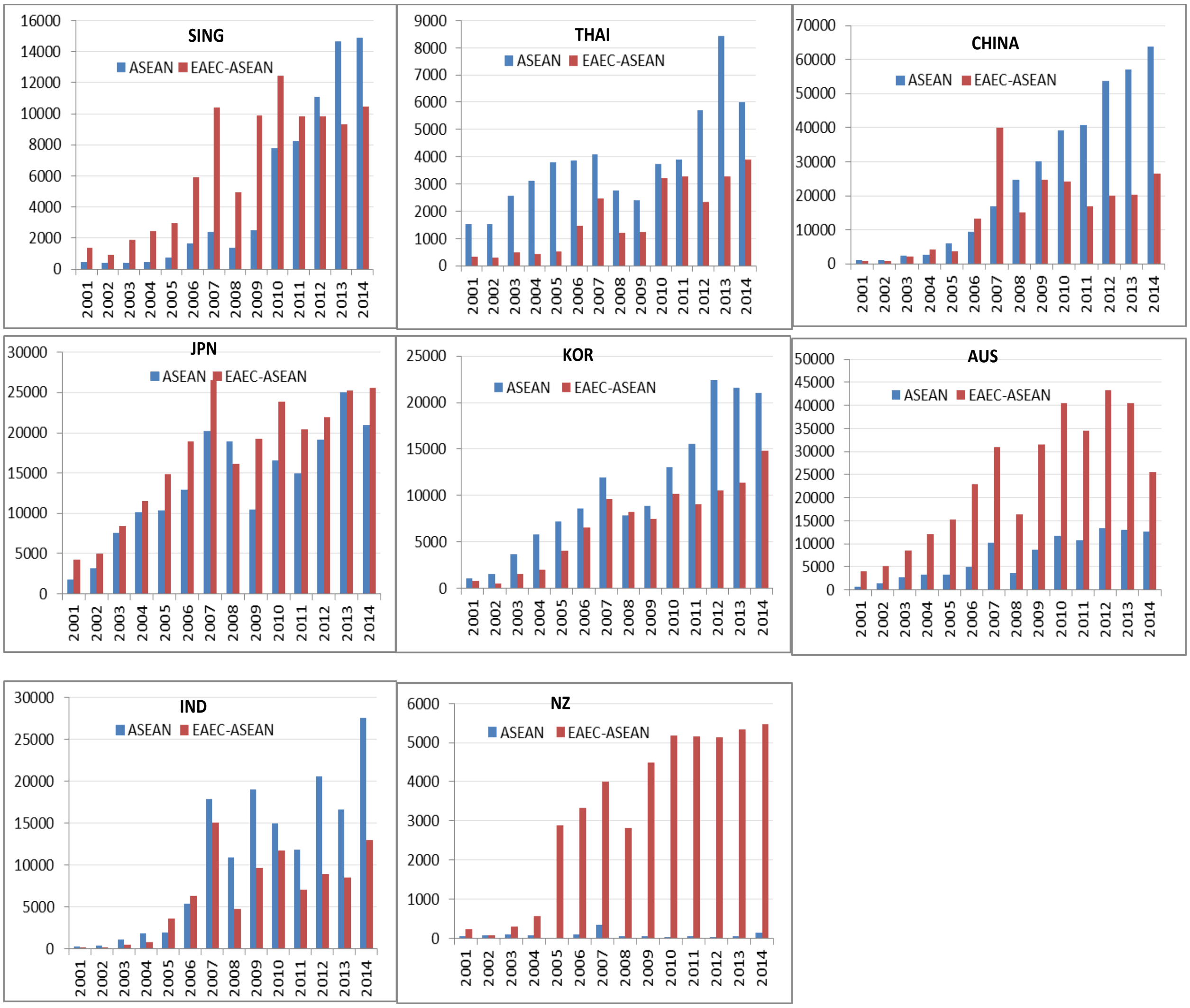
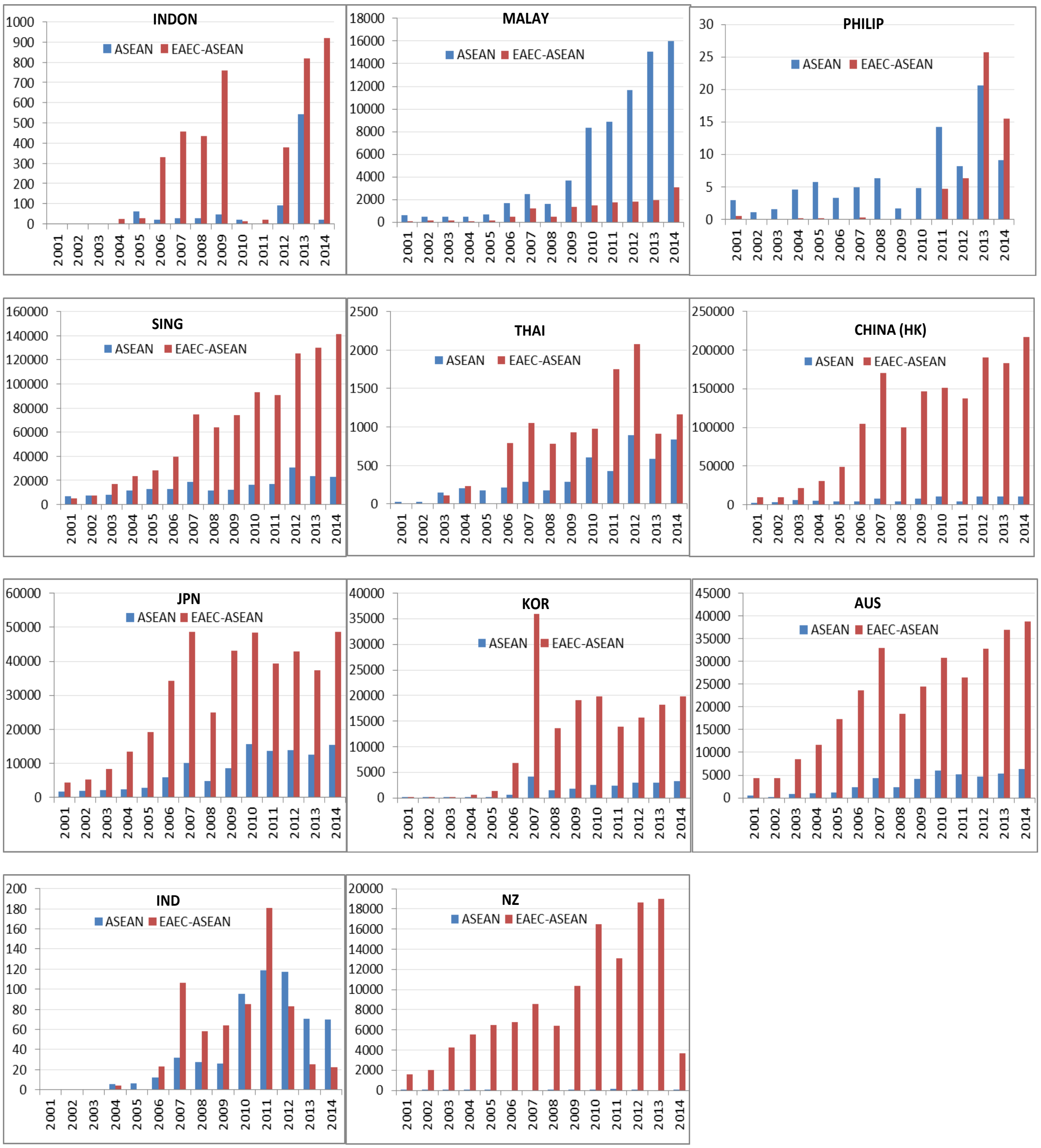


Figure 3 shows the annual values of portfolio equity outflows from sample countries to ASEAN and EAEC-ASEAN groups respectively. There has been a general increase over the years in the quantum of portfolio equity inflows/outflows into/from sample countries from/to ASEAN and EAEC-ASEAN group respectively thereby indicating an increasing level of pan Asian integration.

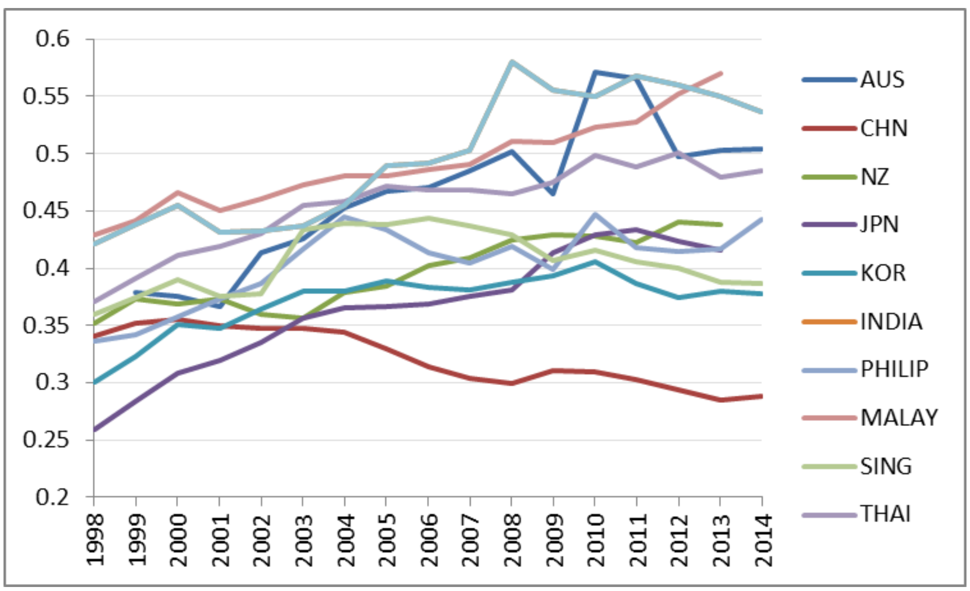
Figure 3: Portfolio Equity Outflows from sample Asian countries into ASEAN and EAEC-ASEAN groups (all figures are end of period and in millions USD)
Source: IMF CPIS, Authors calculations



This process has only strengthened post the global economic crises of 2008 which implies that there is mutual confidence amongst sample Asian countries in fostering cooperation and emerge as cohesive regional economic bloc for withstanding global economic shocks.

The portfolio equity inflows to the ASEAN-5 countries (refer figure 2), with the exception of Singapore (till 2011), was predominantly from the ASEAN than the EAEC-ASEAN group countries. However, when we look at outflow of portfolio equity investments from ASEAN-5 countries (refer figure 3), with the exception of Malaysia and Philippines (till 2012), the quantum of investments seem to be flowing to EAEC-ASEAN countries rather than ASEAN group of countries. Malaysia seems to be the only country in the ASEAN-5 group where both the portfolio equity inflows as well as outflow shave been growing higher with the ASEAN member countries thereby indicating its increasing integration with the ASEAN bloc. This can also be corroborated from the fact that the graph (Refer Figure 4) of regional trade intensity (ratio of total trade with the bloc members to total trade with the world) shows that generally Malaysia has the highest level of regional trade integrationamongst the sample countries under study.

Figure 4: Regional Trade Intensity (ratio of total trade with the bloc members to total trade with the world)



Source: World Bank, Authors Calculations

As for the EAEC-ASEAN group of countries individually, the inflow of portfolio equity investments in China, India and Korea is predominantly on the rise from the ASEAN group of countries than the EAEC-ASEAN group. It appears that though the portfolio equity inflows into the ASEAN-5 member states are mainly from the ASEAN group countries but majority of these big countries in ASEAN are themselves investing in the big developing economies of Asia (and of the world) i.e China, India and Korea. The equity portfolio inflows into Australia and New

Zealand are predominantly from the EAEC-ASEAN group which may be due to its increasing trade linkages with China and Japan. However, when it comes to portfolio equity investment outflows from the EAEC-ASEAN group of countries, they seem to be a directed generally towards EAEC-ASEAN group than ASEAN group with the exception of Japan.

The results from price based analysis based on the static copula models show that the process of pan Asian integration has all but begin with increasing dependence parameter of the EAEC-ASEAN group countries with the Asian benchmark index than that of the ASEAN-5 member countries. Infact, the average dependence parameter of the ASEAN-5 countries, who were the original signatories to the Bangkok declaration, is reduced to almost half by including the other five relatively new member states of ASEAN while computing the average for the whole of ASEAN. Amongst the sample countries, except Singapore, 4 out of the 6 EAEC members (China, Japan, Korea and Australia) have high dependences with the Asian benchmark index. The charts from the dynamic copula models support the preliminary findings of the unconditional models as Japan showed the highest co-movement with the Asian benchmark index over time. Amongst the ASEAN basket, Singapore shows the highest intertemporal dependence with the benchmark index. The dynamics of the dependence measure between the new ASEAN members beyond the original 5 member states with the Asian benchmark is very low. The results based on quantitative indicators reconfirm the process of pan Asian integration as there seems to be an increase in portfolio equity investments inflows/outflows into/from the sample countries from/to ASEAN and EAEC-ASEAN group over time. While the equity inflows into countries comprising ASEAN-5 group are both from the ASEAN and EAEC-ASEAN groups but the outflows from these countries are predominantly flowing to big countries comprising EAC-ASEAN group (China, India, and Korea). Malaysia is the only country in ASEAN where both the portfolio equity inflow/outflow seem to be dominated from the ASEAN bloc itself. However, except Japan, the equity inflows/outflows into/from countries comprising EAEC-ASEAN are dominated by EAEC bloc.

Section 6: Fundamental Determinants of Financial Integration

6.1 Selection of Explanatory Variables

After estimating the time-varying dependence measure for the sample Asian countries under consideration, in this section we investigate the potential explanatory variables driving the stock market integration. We consider the exhaustive list of potential determinants of financial integration that have been considered by various studies in the literature which was summarized in Table 1. These explanatory variables are primarily divided into Macroeconomic, Trade and Market based categories. These variables potentially play an important role in explaining the intertemporal behavior of equity market integration measured by the time varying copula parameter provided by the best model based on goodness of fit test.

6.2 Measurement of Explanatory Variables

These variables are operationalized as can be seen from Table 1. The dependence measure i.e time varying copula parameter of each country was also averaged across time to arrive at a measure to match the observation frequency (annual) of the explanatory variables under study. As is evident, that these explanatory variables under study are measured on different scales hence we first standardize them individually before using them in further analysis. Correlation Analysis[20] showed that some of these explanatory variables were highly associated. Hence, we perform principal components analysis (PCA) for measuring the common variations and to isolate the common components i.e factors[21] in these explanatory variables. The Macroeconomic category contained 6 factors (Refer table 1) namely: Economic Growth and Development (EGD), External Position (EXP), Monetary Policy (MoP), Fiscal Policy (FiP), Governance (GOV) and Economic Risk (Ri). The Trade category was contained 2 factors (Refer table 1): Trade Openness (TO) and Trade Linkages (TL). The Market category was decomposed into 2 factors (Refer table 1) namely: Market Sophistication (MktS) and Market Performance (MktP). We compute time series vectors of these Principal components[22] from their corresponding loadings inferred from the rotated component matrix.

Vo and Daly, 2007; Guesmi et al., 2013; Ng et al., 2013 suggest that growth and development is intricately linked to international financial integration as economic growth acts as a stimulant for private capital flows into the economies thereby fostering higher integration. We expect on a priori ground that a favourable External Position (Guesmi et al., 2013) and Fiscal Position will contribute to equity market integration (Vo and Daly, 2007; Arfaoui and Abaoub, 2010). The Monetary Policy variables act as proxies for confidence shocks on consumption and investment opportunities and thus contribute to valuations of the firms in the equity market (Guesmi et al., 2013). A stable and favourable monetary policy would promote equity market integration. International financial integration is very closely associated with development of a sound institutional, legal and investment environment reflected by the Governance factor (Vo and Daly, 2007). The Risk factor is specified in the international asset-pricing models where they it has been recognized as a factor for financial disintegration (Huoy and Goh, 2007). Integration in equity markets is likely to be driven by increase in Trade Linkages and Trade Openness (Huoy and Goh, 2007; Ng et al., 2013). A developed and well performing financial market (Market Sophistication and Market Performance) helps to attract foreign investors to diversify their portfolio and increase portfolio investment inflows thus contributing to financial integration (Vo and Daly, 2007).

6.3 Model Estimation

This paper uses panel data models to identify the determinants of equity market integration. A panel regression has several advantages 1.) It offers more flexibility in modeling the heterogeneity bounded in the market integration process across different individual equity markets 2.) It reduces collinearity and provides higher degrees of freedom thereby increasing the efficiency of the estimator. Unit Root tests[23] are performed to establish the stationarity of the Principal Components extracted. The basic panel framework is the regression of the form:

$$\rho_{i,t} = \boldsymbol{\alpha} + \sum_{k=1}^{K} \beta_{kit} X_{kit} + \delta_{i,t}$$

$$Y = X \quad \beta \quad + \delta$$

(NT*1) (NT*K) (K*NT) (NT*1)

Where dependent variable is annual dependency measure of country i, at time t. α is a constant and X_{kit} is the kth independent variable (EGD, EXP, MoP, FiP, TL, TO, GOV, Ri, MktP, MktS) of the country i, at year t. Country i= 1,….., 11 and t =1999,……..,2014. $\beta_k$are parameters to be estimated. $\delta_{i,t}$ is an error term that includes cross-section component of the disturbance term i.e μ_i, period effects across observations i.e. θ_t and idiosyncratic error i.e $\varepsilon_{i,t}$ (As $\delta_{i,t} = \mu_i + \theta_t + \varepsilon_{i,t}$)

The cross-section or period effects, however, may not be fixed but can be randomly distributed. To verify whether a random effects model is more superior to the fixed effects model, the specification test constructed by Hausman (1978) is used to test for the orthogonality of the random effects and the independent variables.

6.4. Empirical Results

The large and significant Hausman test statistic (statistic 48.02 with a p value of 0.00) indicates that the fixed effects specification is preferred over random effects specification. The estimates for the two-way fixed effects specification are reported in Table 7.

Table 7: Results of Panel Data Analysis for Determinants of Financial Integration

Independent Variables	Coefficient	Standard Error	T stat
Economic Growth & Development (EGD)	0.0039	0.0039	1.01
External Position (EXP)	0.0022	0.0033	0.65
Monetary Policy (MoP)	0.0002	0.00034	0.05
Fiscal Position (FiP)	-0.0042	0.0039	-1.08
Trade Openness (TO)	.0033*	0.0011	3.1
Trade Linkages (TL)	0.0076**	0.0041	1.89
Governance (GOV)	-0.0046	0.0053	-0.86
Economic Risk (Ri)	0.0014	0.0044	0.31
Market Sophistication (MktS)	0.0024	0.0057	0.43
Market Performance (MktP)	0.1188*	0.0052	2.3
Constant	0.4084*	0.0018	231.5

Notes: */** indicates significance at 5%/10% level of significance

Besides the usual t statistics, robust standard errors are also reported for the LSDV estimates. The White cross-section standard error is robust to cross equation (contemporaneous) correlation as well as different error variances in each cross section. Trade Openness and Market Performance seem to be significant drivers of financial integration. Trade Linkages also seem to contribute to financial integration though weakly (10% level of significance). The influence of trade integration, measured by trade openness and trade linkages, on stock market integration was found to be significant in conformity with the exiting literature (Chin and Forbes, 2004; Wälti, 2011). Local stock market conditions (measured by P/E ratio and Dividend yield of the index) reflect high growth opportunities that help in attracting international investors thereby contributing to market integration (Bekaert et al., 2007).

Section 7: Summary and Policy Suggestions

In this paper, we examine the extent of financial integration in the equity market segments of the ASEAN+6 member countries (group of 10 ASEAN countries and 6 other countries) referred in thisstudy as East Asia Economic Community region. Daily stock market index closing price data was usedfor the 15 countries for the period1 January 1999 (post the Asian financial crises) to 31st March 2015 as data for Brunei was not available. Our empirical analysis is based on copula Garch approach to capture the dynamic process of financial integration by investigating the return co-movement of each sample country's respective equity index with that of the Asian benchmark equity index (used as a proxy for the region). The results obtained show that the level of integration is increasing over time for all countries in the sample study. Further the subsets of sample countries exhibit different levels of financial integration. The highest level of integration is observed for EAEC-ASEAN group followed by ASEAN-5 members. While the integration levels seem to be comparatively low for the full ASEAN region. Referring to individual countries, the level of integration is highest for Japan followed by Korea and China. Amongst the ASEAN member states, Singapore market seems to be the most integrated with the Asian benchmark. Our quantity based measures reconfirm the findings of price based analysis, as shown by greater equity portfolio flows between the sample countries and the ASEAN and EAEC-ASEAN groups.The major portfolio inflows in the ASEAN-5 countries are from the ASEAN region, while the ASEAN-5 portfolio outflows are mainly directed to EAEC-ASEAN region. In contrast, both portfolio inflows as well as outflows for the EAEC-ASEAN countries are predominantly within their group. The panel data analysis was performed to identify the fundamental determinants of financial integration in the Asian region. We find that trade integration (as measure by trade openness and trade linkages) and domestic stock market performance are thekey drivers of inter temporal dependences.

As demonstrated by Capannelli et al. (2009) and Yu et al. (2010), in order to increase financial integration within the region, it will be necessary to strengthen the overall financial stability of the region, to promote macroeconomic policy coordination and to develop regional institutions. In the backdrop of the formation of the ASEAN Economic Community in 2015, based on our results, we suggest extending the ASEAN bloc to a more viable ASEAN+6 bloc thereby eventually forming an East Asian Economic Community region in the future. The possible formation of a bigger and more representative East Asian Economic Community will help the region to play a more effective role in shaping a world trade and financial system that is more responsive to its needs and will be the cornerstone to what is described as the “Asian Century”. The East Asian region includes some of the fastest growing economies in the world. The EAEC region constitutes a huge market and hence could form a vibrant regional bloc that would be roughly of the size of the EU in terms of GDP, will have larger magnitude of trade than NAFTA and international reserves bigger than those of EU and NAFTA put together (Kumar, 2004). Regional integration will be fostered as the policymakers of the region prepare to strengthen their macroeconomic fundamentals, enhance trade policy cooperation, undertake capital market development, effectively manage cross-border portfolio investments, and strengthen the regional institutional frameworks. Taking cues from Europe Union, the ASEAN group members outside of ASEAN-5 may be asked to increase their trade integration and capital market development for strengthening their integration in to the EAEC. Similarly, the big economies in the region- Japan, China, Korea, and India should take the lead to facilitate the process of integration.

The study has important implications for the policy makers, portfolio managers and academia. For the global policy makers, the study is of particular relevance in the light of growing number of regional co-operation initiatives worldwidefor achieving political influence, enhanced linkages amongst economies as well asdeveloping a coordinated response to global risks including financial contagion. Further varying degree of financial integration exhibited by the EAEC subgroups provides investment opportunities and risk diversification possibilities for global portfolio managers. Some of the EAEC economies are expected to be large enough in near future with a possibility of dedicated country funds. Greater economic and trade integration and better capital market development by these countries shall accelerate this process. From the academic point of view, the study highlights the dynamic nature of the financial integration process and identifies the fundamental determinants to achieve this goal. The paper contributes to the literature by measuring the progress of financial integration in the equity market segments for the ASEAN+6 members’ countries. It also provides strong arguments in expanding ASEAN Economic Community region into a more economically viable EAEC region.

Notes:

1. According to Asian Economic Integration Monitor (AEIM) launched by the Asian Development Bank (ADB) in July 2012, regional integration is a process leading to greater interdependence in such areas as trade, investment and finance

2. Financial integration is achieved when all economic agents in the financial markets face identical rules and have equal access to financial instruments or services in these markets and are treated equally. (Baele et al., 2004).

3. Financial market integration creates powerful internal pressures for financial reform and development by encouraging further financial liberalization and upgrading of financial capacity thereby contributing to economic growth. See (Levine, 2001), (Gianetti et al., 2002) (Baele et al., 2004), (De Brouwer and Corbett, 2005), (Calvi, 2010) for further details.

4. Unilateral liberalized approach is whereby country makes tariff reductions independently and without reciprocal action by other countries. Source: International Trade Agreements by Douglas A Irwin http://www.econlib.org/library

5. ASEAN was created with the signing of Bangkok Declaration by Indonesia, Malaysia, Philippines, Singapore, and Thailand. Subsequently the ASEAN bloc grew with the addition of Brunei, Vietnam, Laos, Myanmar and Cambodia (in the order of their entry). The 10 ASEAN members plus the 3members i.e Japan, China and South Korea constitutes the ASEAN+3 bloc. The 2nd East Asian Summit (EAS) was held Cebu in January 2007 wherein ten ASEAN members and six countries including China, Japan, South Korea, India, Australia, and New Zealand participated. Japan regards this ASEAN+6 (EAS group) as an appropriate group for East Asia's trade and investment cooperation.

6. Financial Integration in a region is a strategic choice to enhance trade and firm cooperation within a geographical area (Boubakri and Guillaumin, 2015)

7. Economic integration is the unification of economic policies between different countries through the partial or full abolition of tariff and non-tariff restrictions on trade taking place among them prior to their integration. Source: Wikipedia

8. Economic and financial integration are mutually enforcing each other while empirical evidence seems to suggest that economic integration provides a channel for financial integration (Phylaktis and Ravazzolo, 2001).

9. Feldstein and Horioka (1980) measure the degree of financial integration using the correlation of domestic saving with investment rates

10. For a survey of the literatures and various indicators, see (Cavoli et al., 2004); (Baele et al., 2004); (Poonpatpibul et al., 2006).

11. For survey of literature using these indicators, see (Yu et al., 2010; Narayan et al., 2014; Boubakri and Guillaumin, 2015; Chien et al., 2015, Gupta et al.,2015)

12. Refer Federal Reserve History of timeline of the Asian Financial Crises (July 1997- December 1998). www. federalreservehistory.org/Events/DetailView/51

13. MSCI indices were selected as they are value weighted indices and have a broad coverage of markets in terms of market capitalization and are compiled with similar criteria and hence are homogenous for cross country comparison. They are thus widely employed in literature.

14. EAEC- ASEAN group members include China, Japan, South Korea, India, Australia and New Zealand

15. The global category containing variables such as world interest rate, liquidity, industrial production, growth opportunities, index returns, oil/gold price changes, market/term/default premium, and volatility of world returns have been omitted in our analysis owing to the fact that they do not vary for our cross section data (countries) and hence pose estimation problem while employing panel data technique. However, we believe that our global market proxy (US returns) used in the AR(1) based mean equation accounts for the impact of these factors on the country returns.

16. Annualization has been done assuming 250 trading days in an year

17. Though Myanmar became an independent nation in 1948, its military dictatorship rule formally ended in 2011 thereby leading to not much development of its equity markets

18. Unit root test results are not reported due to brevity of space. These are available on request from the authors.

19. Mashal et al. (2003) show that the empirical fit of the t-copula is generally superior to that of the Gaussian copula.

20. Results of Correlation tests are not reported due to brevity of space. These are available on request from the authors.

21. The factors have been named in accordance with the underlying explanatory variables constituting the same

22. Results of the Principal Component tests including rotated component matrix and coefficient matrix are not reported due to brevity of space. These are available on request from the authors.

23. Unit root tests for panel data are not reported here and are available on request from the authors

Acknowledgments

The authors would like to thank Dr. Osvaldo Candido da Silva Filho for his MATLAB copula package which helped us to run our empirical tests.

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